

BOUNDARY CONDITIONS FOR THE ELECTROMAGNETIC FIELDS VECTORS: B , E , D , AND H , (AT THE INTERFACE BETWEEN TWO MEDIA)

In order to discuss the behaviours of electromagnetic waves at the boundary.

We shall now investigate the boundary conditions which the time dependent electromagnetic field vectors $\vec{B}$, $\vec{E}$, $\vec{D}$ and $\vec{H}$, must satisfy at the interface between two different media. Boundary conditions are:

1. The normal component of magnetic induction $\vec{B}$ is **continuous** across boundary, i.e.,

$$B_{1n} = B_{2n}.$$

2. The tangential component of $\vec{E}$ is **continuous** across the interface, i.e.,

$$E_{1t} = E_{2t}.$$

3. The normal component of electric displacement $\vec{D}$ is **discontinuous** across the interface, i.e.,

$$D_{1n} - D_{2n} = \sigma.$$

4. The tangential component of magnetic induction $\vec{H}$ is **continuous** across the surface separating two dielectrics, i.e.,

$$H_{1t} = H_{2t}.$$

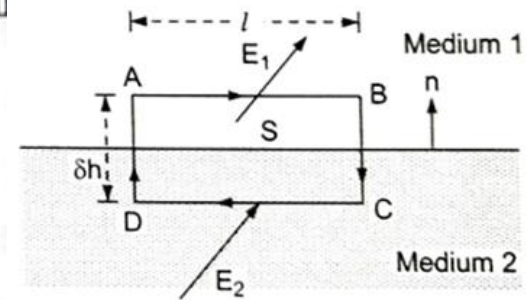


Fig. 12.5 Rectangular loop at the interface between two media

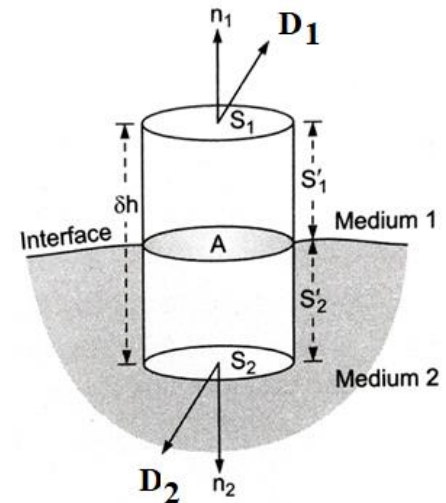


Fig. 12.4 Pill box shaped surface at the interface of two media.

Prove that The tangential component of $\vec{E}$ is *continuous* across the interface, i.e., $E_{1t} = E_{2t}$.

Q. Develop boundary conditions for $\vec{E}$ separated by two dielectric media of different permittivity

Boundary Condition for $\vec{E}$

The boundary condition, which the tangential component of electric field must satisfy, may be obtained from Maxwell's equation

$$\text{curl } \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \dots(67)$$

At the interface, we construct a rectangular loop $ABCD$ bounding a surface S as shown in Fig. (12.5). Integrating eqn. (67) over the surface bound by rectangular loop $ABCD$, we have

$$\int_S \text{curl } \vec{E} \cdot \vec{n} dS = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{n} dS \quad \dots(68)$$

Transforming the surface integral of the left hand side into a line integral over the path $ABCD$ with the help of Stoke's theorem, eqn. (68) yields.

$$\begin{aligned} \int_{ABCD} \vec{E} \cdot d\vec{l} &= \int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{n} dS \\ \text{or } \int_{AB} \vec{E}_1 d\vec{l} + \int_{CD} \vec{E}_2 d\vec{l} + \text{contribution from sides } BC \text{ and } DA &= - \int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{n} dS \quad \dots(69) \end{aligned}$$

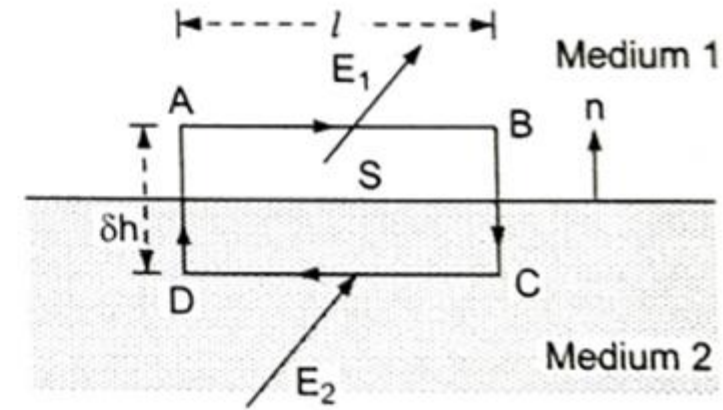


Fig. 12.5 Rectangular loop at the interface between two media

$$\text{or } \int_{AB} \vec{E}_1 d\vec{l} + \int_{CD} \vec{E}_2 d\vec{l} + \text{contribution from sides } BC \text{ and } DA = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot \vec{n} dS \quad \dots(69)$$

If the loop is now shrunk (letting $\delta h \rightarrow 0$), the contribution from sides BC and DA will vanish. Provided $\partial \vec{B} / \partial t$ is everywhere finite, the right hand side of eqn. (69) also vanishes because the area of integrating also vanishes *i.e.*, no flux can be enclosed. Thus when $\delta h \rightarrow 0$.

$$\int_{AB} \vec{E}_1 d\vec{l} + \int_{CD} \vec{E}_2 d\vec{l} = 0$$

$$\text{or } \vec{E}_1 \vec{AB} + \vec{E}_2 \vec{CD} = 0$$

$$\text{or } \vec{E}_1 \vec{AB} - \vec{E}_2 \vec{AB} = 0$$

$$\text{or } \vec{E}_{1t} = \vec{E}_{2t} \quad \dots(70)$$

where E_{1t} and E_{2t} are the tangential components of the electric field in the two media. Equation shows that the ***tangential components of E are continuous across the interface.***

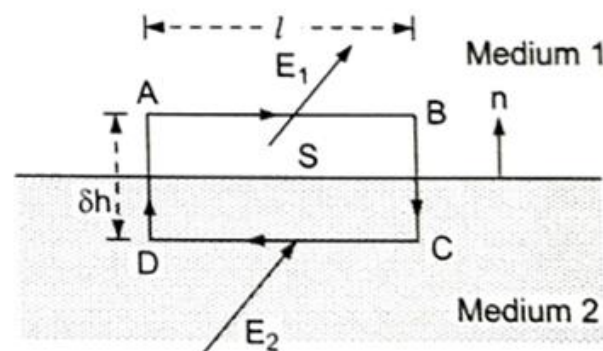


Fig. 12.5 Rectangular loop at the interface between two media

Prove that The normal component of electric displacement $\vec{D}$ is *discontinuous* across the interface, i.e., $D_{1n} - D_{2n} = \sigma$.

Q. Develop boundary conditions for $\vec{D}$ separated by two dielectric media of different permittivity

Boundary Condition of Electric Displacement $\vec{D}$

The boundary condition for the electric displacement $\vec{D}$ can be obtained from Maxwell's equation

$$\text{div } \vec{D} = \rho. \quad \dots(71)$$

Integrating eqn. (71) over the pill box shaped volume V of Fig. 12.4, we obtain

$$\int_V \text{div } \vec{D} dV = \int_V \rho dV \quad \dots(72)$$

Transforming the volume integral of left hand side of eqn. (72) in surface integral with the help of Gauss divergence theorem, we get

$$\oint_S \vec{D} \cdot \vec{n} dS = \int_V \rho dV$$

$$\text{or } \int_{S_1} \vec{D}_1 \cdot \vec{n}_1 dS + \int_{S_2} \vec{D}_2 \cdot \vec{n}_2 dS + \text{contribution from walls} = \int_V \rho dV$$

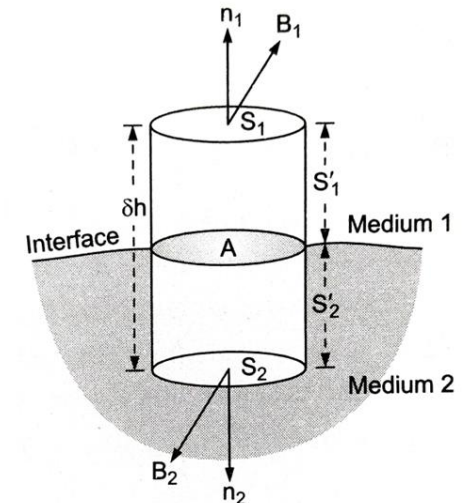


Fig. 12.4 Pill box shaped surface at the interface of two media.

Now let us suppose that the surfaces S_1 and S_2 approach each other towards the interface so that $S_1 = S_2 = A$, i.e., $\delta h \rightarrow 0$ the contribution from walls tends to zero. Now, instead of volume charged density ρ , the concept of surface charge density σ must be used, i.e.,

$$\lim_{\delta h \rightarrow 0} \int_V \rho dV = \int_V \rho dA = \sigma A$$

$$\int_{S_1} \vec{D}_1 \cdot \vec{n}_1 dS + \int_{S_2} \vec{D}_2 \cdot \vec{n}_2 dS = \sigma A$$

or

$$\vec{D}_1 \cdot \vec{n}_1 S_1 + \vec{D}_2 \cdot \vec{n}_2 S_2 = \sigma A$$

or

$$\vec{D}_1 \cdot \vec{n}_1 + \vec{D}_2 \cdot \vec{n}_2 = \sigma \text{ as } S_1 = S_2 = A$$

or

$$\vec{D}_1 \cdot \vec{n}_1 + \vec{D}_2 \cdot \vec{n}_2 = \sigma$$

or

$$\boxed{D_{1n} - D_{2n} = \sigma}$$

where D_{1n} and D_{2n} are components of the electric displacement in the two media normal to the interface in the direction of $\vec{n}_1$. Evidently, *the normal component of electric displacement is not continuous across the interface and changes by an amount equal to the free surface density of charge at the interface.*

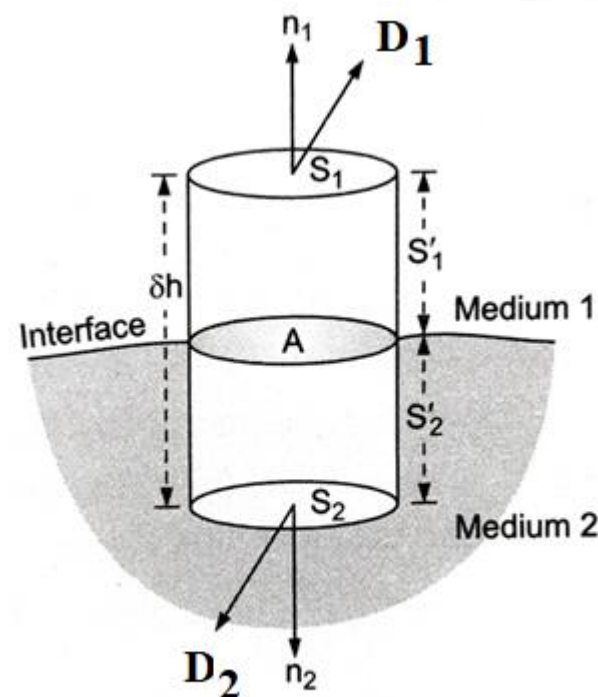


Fig. 12.4 Pill box shaped surface at the interface of two media.

DIELECTRIC POLARIZATION

Consider a dielectric bar containing polar or nonpolar molecules. Let us place the bar in an external electric field, say between the plates of a charged capacitor. The presence of the external electric field can cause the shifting of the centres of gravity of the positive and negative charges to convert the non-polar molecules into polar molecules. The result will be the alignment of dipoles along the electric field resulting in a net dipole moment of the dielectric bar Fig. 11.25. The *presence of* positive and negative charges, in the neighbourhood of each other, will cancel the effect of each other within the body of the dielectric. But on the surfaces of the bar at right angles to the external field $\vec{E}_0$, there will appear a net positive and negative charge as shown in Fig. 11.25. The appearance of the net positive and negative charges on the surfaces of the bar will set up an electric field $\vec{E}_p$ within the dielectric in a direction opposite to the applied field. And net electric field within the dielectric will be the vector sum of $\vec{E}_0$ and $\vec{E}_p$, i.e., $\vec{E} = \vec{E}_0 + \vec{E}_p$. As a result, the resultant electric intensity inside the dielectric is diminished, i.e., $E < E_0$ therefore, the actual field acting on the molecules or atoms of a dielectric is E instead of E_0 .

The phenomenon of appearance of the charges on the surfaces of the dielectric bar induced by an external electric field is called polarization. Since dipoles align along the electric field, thus, the process of dipole alignment in a dielectric is also known as polarization. Quantitatively, the polarization P is measured as net electric dipole moment per unit volume, i.e., $P = \frac{\Sigma p}{V}$, where p is the average dipole moment induced on each molecule of a dielectric having volume V . The charge on the plates of the capacitor is termed as free charge and the charge induced on the surfaces of the dielectric is called bound charge.

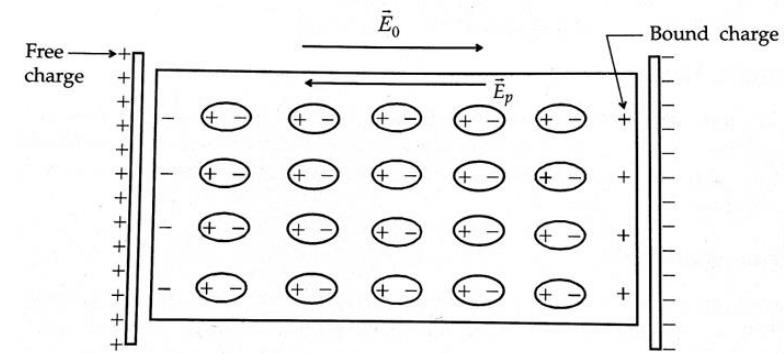


Fig. 11.25 Alignment of Atomic Dipoles when Dielectric Slab is Placed in an Electric Field $\vec{E}_0$

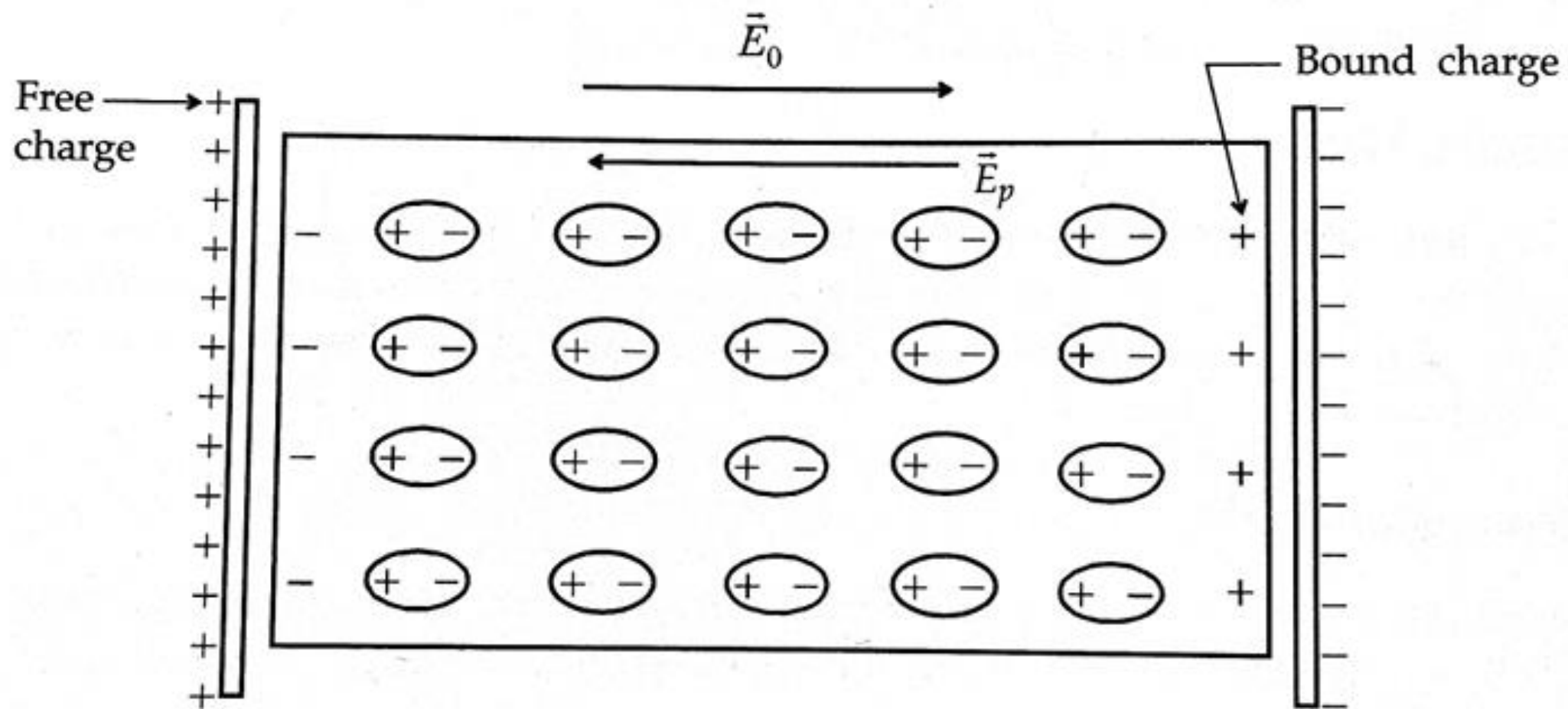


Fig. 11.25 Alignment of Atomic Dipoles when Dielectric Slab is Placed in an Electric Field $\vec{E}_0$

Q4 (b) Prove the relation $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$.

Establish the RELATION BETWEEN $\vec{D}$, $\vec{E}$ AND $\vec{P}$ (THREE ELECTRIC VECTORS)

Q. Deduce the relationship among the three vectors D , E , P in the metre-kilogram-second (mks) or SI system is: $D = \epsilon_0 E + P$ (ϵ_0 is a constant, the permittivity of a vacuum)

RELATION BETWEEN $\vec{D}$, $\vec{E}$ AND $\vec{P}$ (THREE ELECTRIC VECTORS)

Now we will derive the relationship among three electric vectors D , E and P . Let us consider a free charge q .

According to Gauss's law the total electric flux due to this charge will be

$$\oiint \vec{E}_0 \cdot d\vec{A} = q/\epsilon_0 \quad \Rightarrow \quad \oiint \epsilon_0 \vec{E}_0 \cdot d\vec{A} = q$$

or $\epsilon_0 E_0 A = q$

$$\Rightarrow E_0 = \frac{q}{\epsilon_0 A} \quad \dots(49)$$

where E_0 is the electric field due to free charge q . When dielectric is present, induced charge on surface appears, say $-q'$, so that Gaussian surface encloses $q - q'$ charge. Gauss's law then becomes

$$\oiint \epsilon_0 \vec{E} \cdot d\vec{A} = q - q'$$

$$\oiint \epsilon_0 \vec{E} \cdot d\vec{A} = q - q'$$

or

$$\epsilon_0 EA = q - q'$$

$\Rightarrow$

$$E = \frac{q - q'}{\epsilon_0 A} = \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A}$$

or

$$\frac{q}{\epsilon_0 A} = E + \frac{q'}{\epsilon_0 A}$$

or

$$\frac{q}{A} = \epsilon_0 E + \frac{q'}{A} = \epsilon_0 E + \sigma_p \quad \dots(49a)$$

From eqn. (49), we have $\Rightarrow E_0 = \frac{q}{\epsilon_0 A} \quad \dots(49)$

$$\frac{q}{A} = \epsilon_0 E_0,$$

Also

$$\epsilon_r = \frac{E_0}{E}$$

$\therefore$

$$\frac{q}{A} = \epsilon_0 \epsilon_r E = \epsilon E = D \quad \dots(49b)$$

$$\epsilon_r = \frac{\text{Capacity with dielectric material}}{\text{Capacity with vacuum}} = \frac{\epsilon A/d}{\epsilon_0 A/d}$$

$\Rightarrow$ relative permittivity $\epsilon_r = \frac{\epsilon}{\epsilon_0}$

Therefore, from eqn. (49a) and (49b), we have $D = \epsilon_0 E + \sigma_p$

$$D = \epsilon_0 E + P$$

$$(\because \sigma_p = P) \quad \dots(50)$$

The electric dipole moment will be $\sigma_p Al$, where Al represent the volume of the dielectric bar. If P be the σ_p . *dielectric polarization*, then electric dipole moment is PAI . Therefore, we have

$$PAI = \sigma_p \cdot Al \Rightarrow P = \sigma_p$$

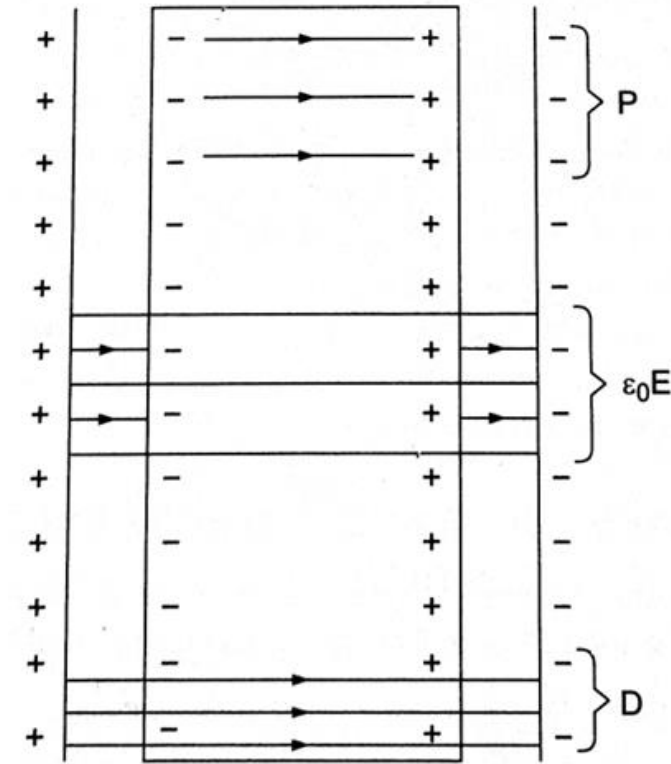


Fig. 11.30

Figures Shows Lines Associated with D (free Charges), $\epsilon_0 E$ (all Charges) and P (Polarization Charges).

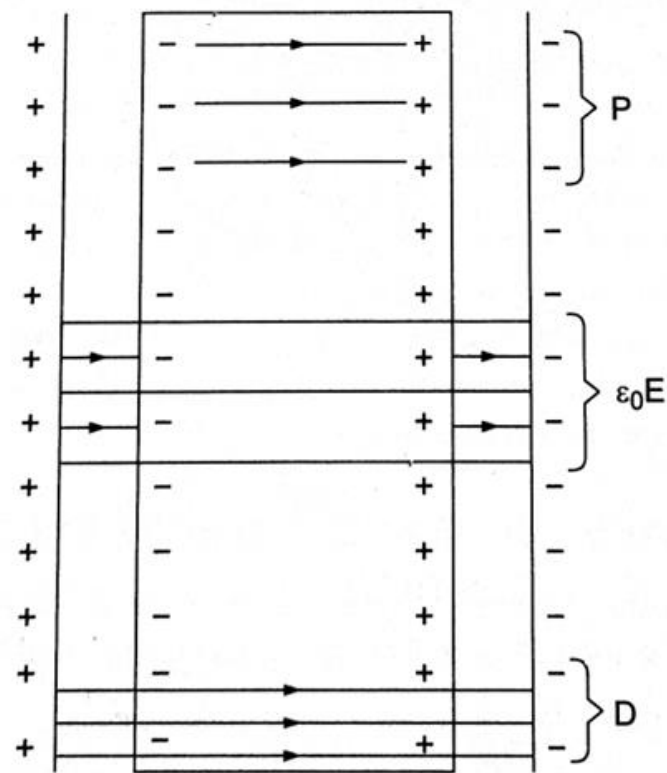


Fig. 11.30 Figures Shows Lines Associated with D (free Charges), $\epsilon_0 E$ (all Charges) and P (Polarization Charges).

It is noted here that D is connected with free charges only and P is connected with polarization charges only but E is connected with both free and polarization charges.

Figure shows lines of D , P and E

where lines of D Begin and end on the free charges, lines of P begin and end on the polarization (bound) charges and lines of E reflect the presence of both types of charges.

Q. Prove that the Electric field at a point is defined as the gradient of the potential at that point.

OR

Q. Formulate the relation between E and V (or U) in differential form

RELATIONSHIP BETWEEN ELECTRIC FIELD (**E**) AND ELECTRIC POTENTIAL (**U or V**)

We have seen that the electric potential is given by

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

Now take gradient of above expression, we have

$$\vec{\nabla} U = \vec{\nabla} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) = \frac{1}{4\pi\epsilon_0} q \vec{\nabla} \left(\frac{1}{r} \right)$$

Since, $\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$, and $r = (x^2 + y^2 + z^2)^{\frac{1}{2}}$

$$\text{we have } \vec{\nabla} \left(\frac{1}{r} \right) = \vec{\nabla} (x^2 + y^2 + z^2)^{-\frac{1}{2}} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2)^{-\frac{1}{2}}$$

$$= -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (\hat{i} 2x + \hat{j} 2y + \hat{k} 2z)$$

$$= -r^{-3} \vec{r} = -\frac{1}{r^2} \frac{\vec{r}}{r} = -\frac{\hat{r}}{r^2} \quad (\because \vec{r} = |\vec{r}| \hat{r})$$

$$\therefore \vec{\nabla} U = \frac{1}{4\pi\epsilon_0} q \left(\frac{-\hat{r}}{r^2} \right) = -\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

or

$$\vec{\nabla} U = -\vec{E}$$

$$\left(\because \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \right)$$

or

$$\vec{E} = -\vec{\nabla} U$$

...(3)

Thus, electric field at a point is defined as the gradient of the potential at that point.

Q.3(a) Show that electrostatic field is conservative in nature.

Suppose a charge Q is situated in an electric field at points A and B . The path from A to B is a closed path. The electric field does work to move a test charge from point A to B . To determine the work done, we sum up the work done for all the small segments that the path A to B is divided into. This kind of integral evaluated along a line is known as a line integral.

The line integral of the electric field along A to B

$$\int_A^B \vec{E} \cdot d\vec{l}$$

The line integral of the electric field along B to A

$$\int_B^A \vec{E} \cdot d\vec{l}$$

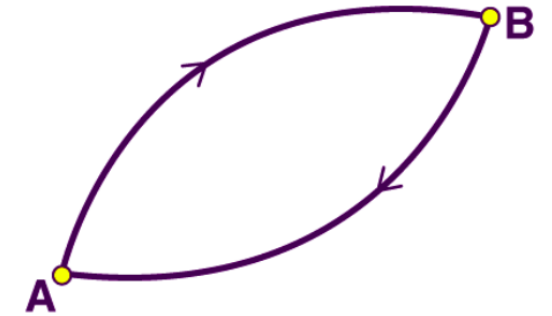
Hence, the line integral along the closed path as shown in the figure above is given by

$$\int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l}$$

But,

$$\int_B^A \vec{E} \cdot d\vec{l} = - \int_A^B \vec{E} \cdot d\vec{l}$$

$$\int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l} = \int_A^B \vec{E} \cdot d\vec{l} + - \int_A^B \vec{E} \cdot d\vec{l} \Rightarrow \int_A^B \vec{E} \cdot d\vec{l} + \int_B^A \vec{E} \cdot d\vec{l} = 0$$



Question:- Show that the electrostatic field E is conservative in nature

This demonstrates that the line integral of an electric field along a closed path is zero.

Therefore, the work done by the electric field is independent of the path, meaning it relies only on points A and B .

This confirms that the electric field is indeed conservative in nature.

